Automated Land Cover Classification using Satellite Imagery

(CNN-based Classification with Image Enhancement & Feature Analysis)

Disciplinary Work for Computer Vision & Image Processing

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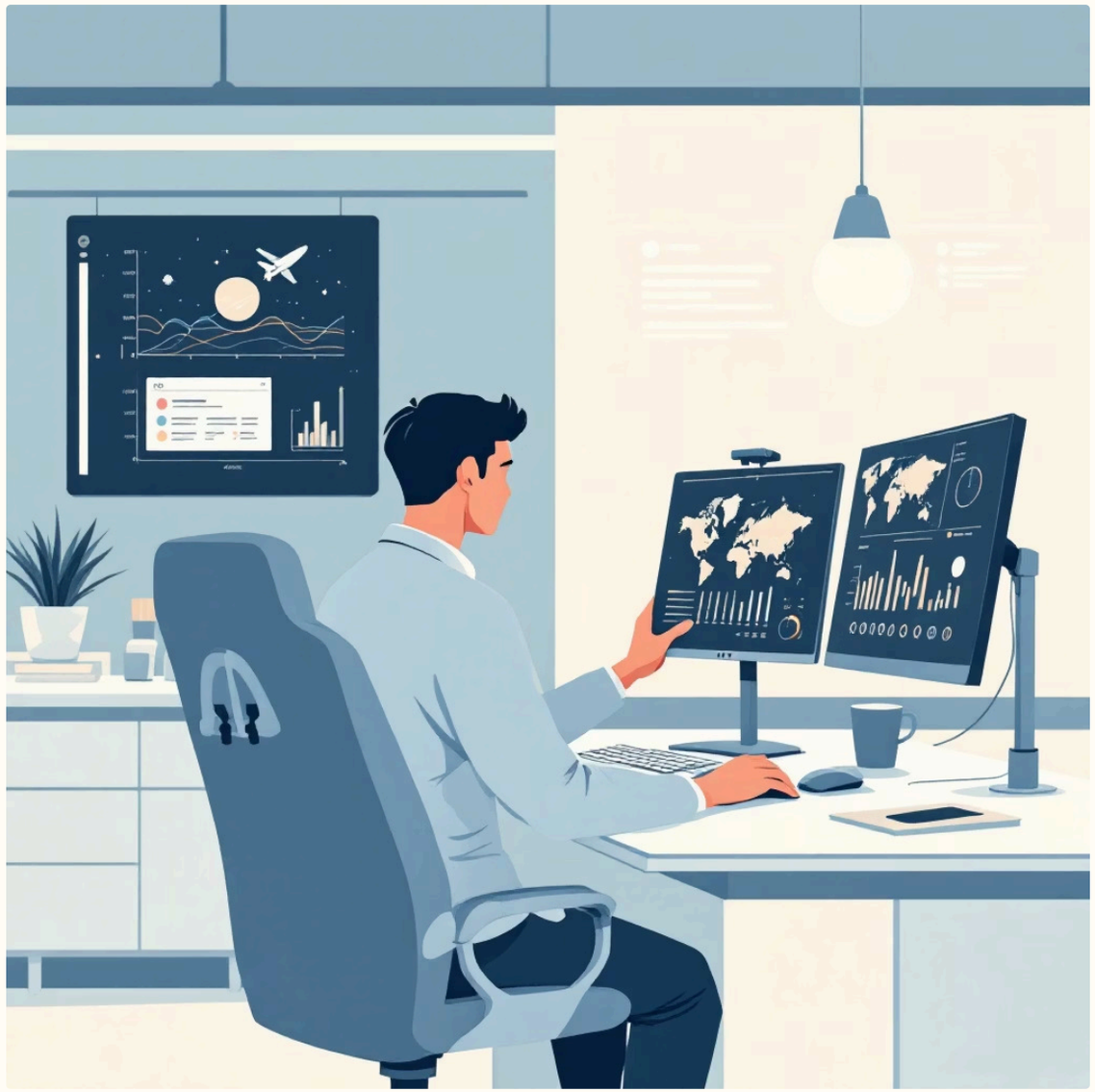
Introduction & Motivation

The Challenge

Manual analysis of high-resolution satellite imagery is labour-intensive, time-consuming, and prone to human error. With satellites generating terabytes of geospatial data daily, automated classification techniques have become essential for efficient Earth observation.

This research addresses a critical gap in **Computational Techniques for Geospatial Data Analysis**, leveraging deep learning to transform raw satellite imagery into actionable intelligence.



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Project Objectives

Digital Image Processing Pipeline

Implement comprehensive preprocessing techniques including enhancement, normalisation, and quality assessment for satellite imagery.

Spectral & Texture Feature Extraction

Extract discriminative features using VARI indices and GLCM texture descriptors to capture spectral and spatial information.

CNN-based Multiclass Classification

Develop and train a deep convolutional neural network for accurate classification across 10 distinct land cover categories.

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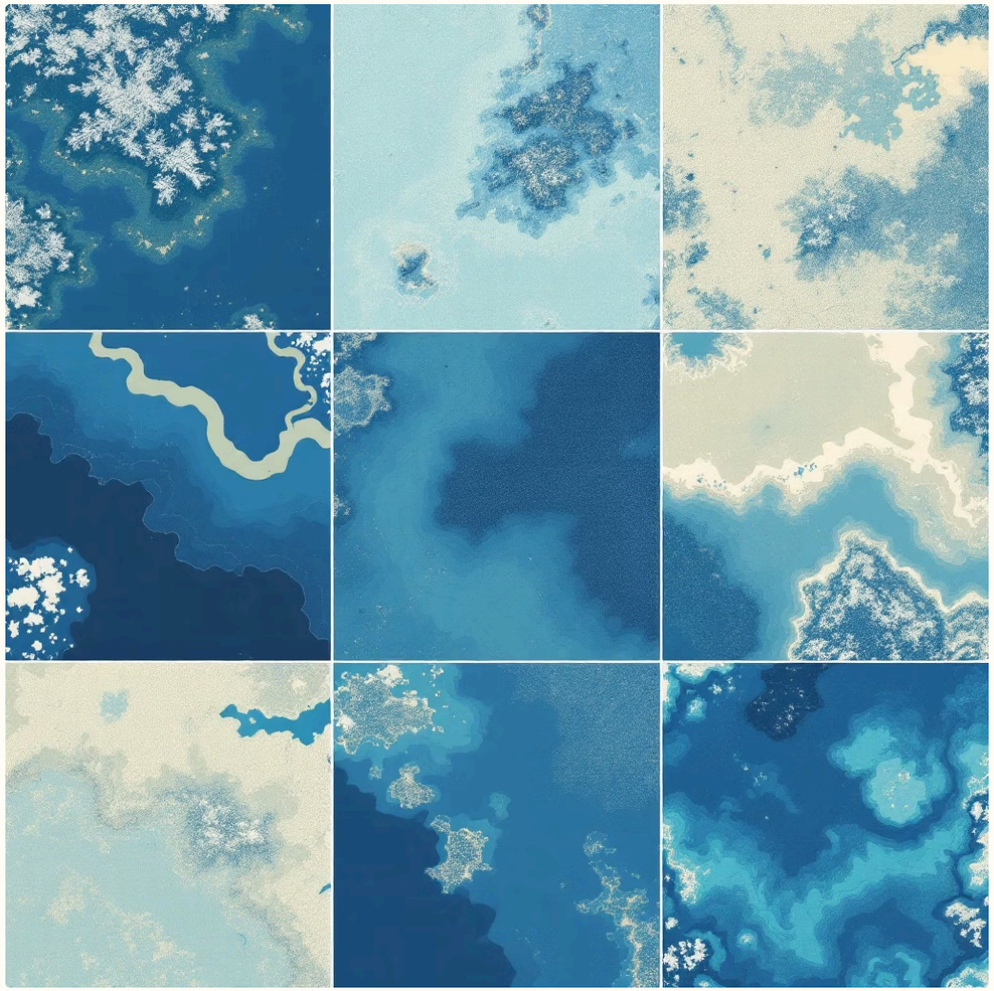
Dataset Overview: EuroSAT

Sentinel-2 Satellite Data

The EuroSAT dataset comprises **27,000 labelled images** derived from the Sentinel-2 satellite mission, covering diverse European landscapes. Each image is a **64 × 64 pixel RGB tile** captured at 10-metre spatial resolution.

10 Land Cover Classes

- Annual Crop & Permanent Crop
- Forest, Herbaceous Vegetation
- Highway, Industrial, Pasture
- Residential, River, Sea & Lake



This balanced dataset enables robust training and evaluation of classification models across varied terrain types and land use patterns.

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Methodology: Preprocessing Pipeline

Topics 1 & 2: Image Enhancement Techniques

Raw Satellite Imagery

64×64 RGB tiles with varying contrast and illumination conditions from Sentinel-2

Histogram Equalisation

OpenCV-based contrast enhancement to improve visual quality and feature visibility

Pixel Normalisation

Scaling pixel values to [0, 1] range for mathematical consistency and faster convergence

These preprocessing steps ensure optimal input quality for subsequent feature extraction and classification stages.

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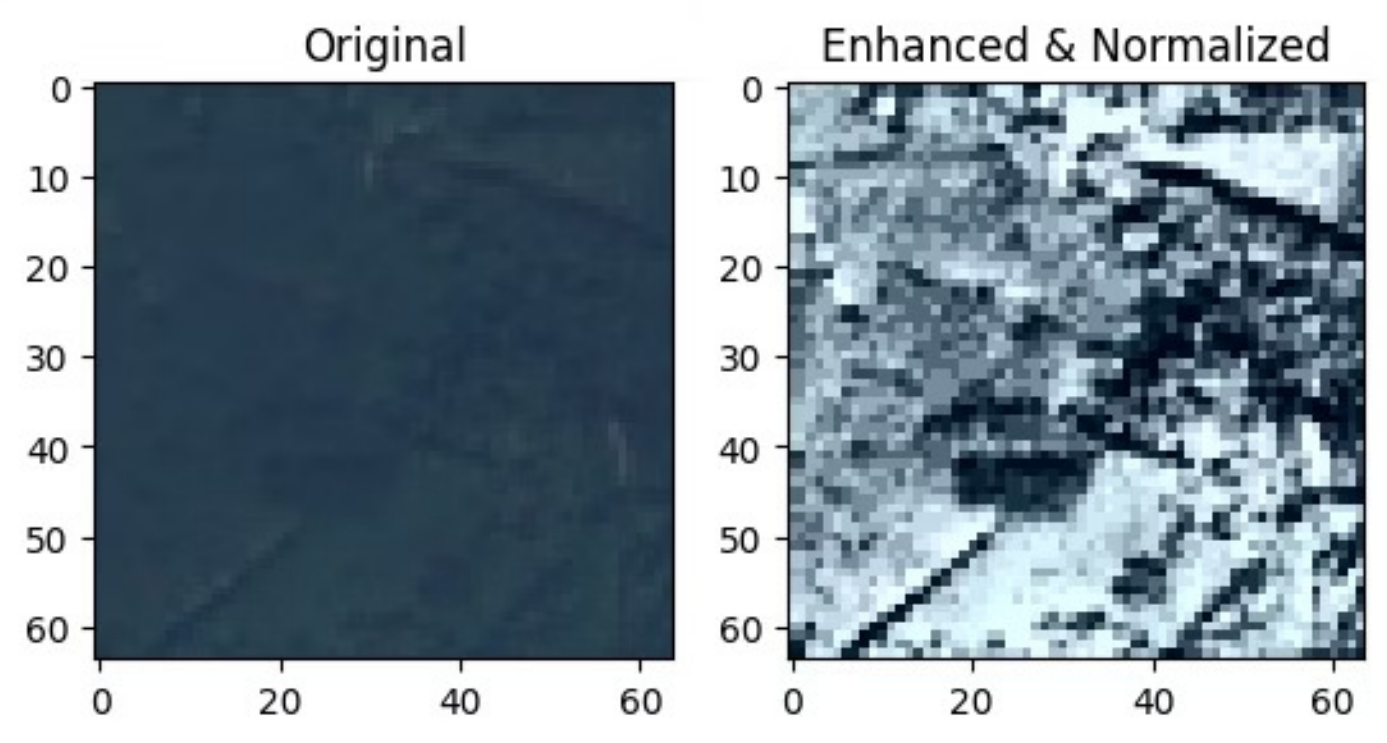
Image Preprocessing Output



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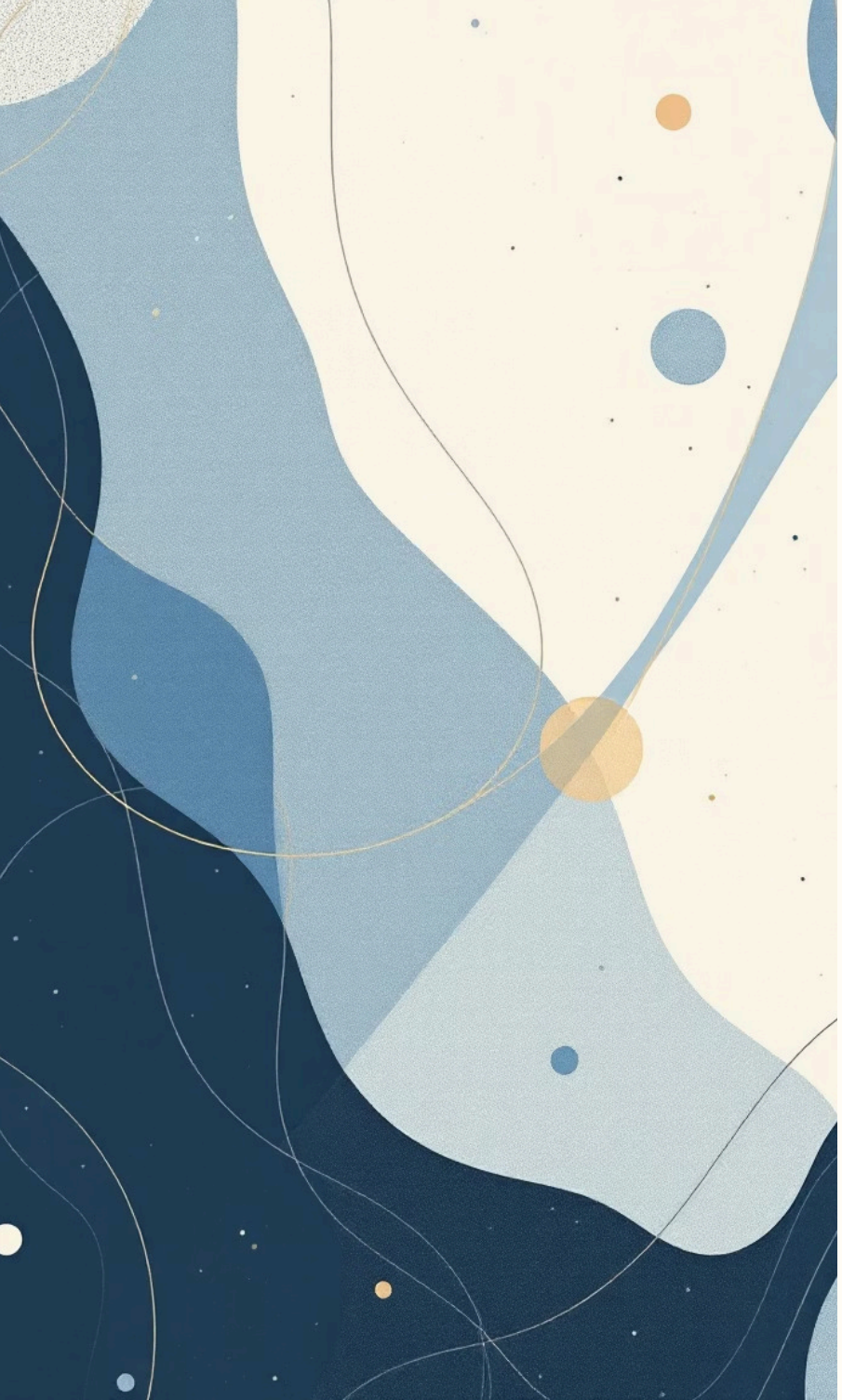

Feature Extraction Techniques

Topics 4 & 5: Capturing Spectral and Spatial Information

Spectral Features: VARI Index

The **Visible Atmospherically Resistant Index** quantifies vegetation presence using spectral band ratios:

$$\text{VARI} = \frac{\text{Green} - \text{Red}}{\text{Green} + \text{Red} - \text{Blue}}$$

This index effectively distinguishes vegetated areas from urban and water bodies while minimising atmospheric interference.

Texture Features: GLCM Properties

The **Grey-Level Co-occurrence Matrix** captures spatial texture patterns through statistical measures:

- **Contrast:** Measures intensity variation (surface roughness)
- **Homogeneity:** Quantifies uniformity (texture smoothness)

These properties distinguish between similar spectral signatures with different spatial structures.

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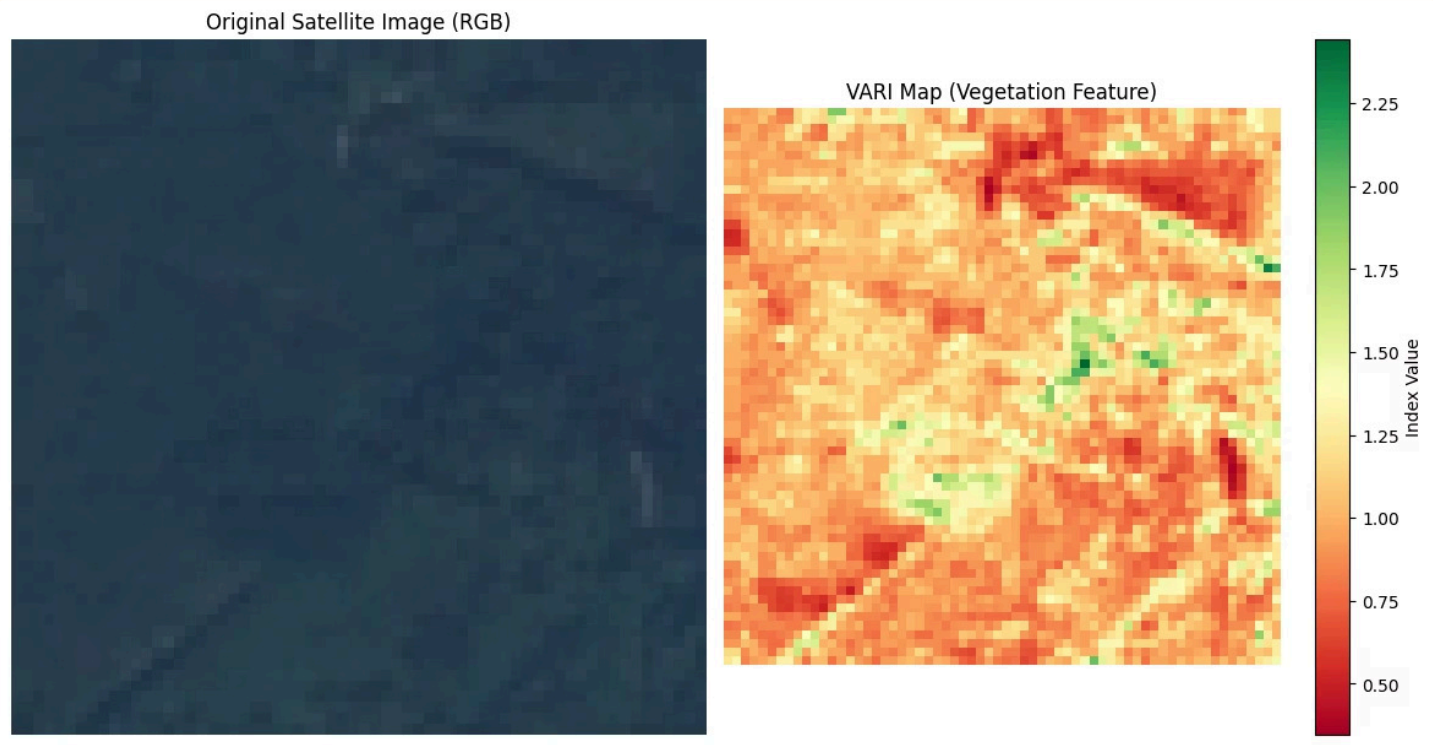
Feature Extraction Output (VARI)



- Original Satellite Image
- Enhanced & Normalized Image (side-by-side)

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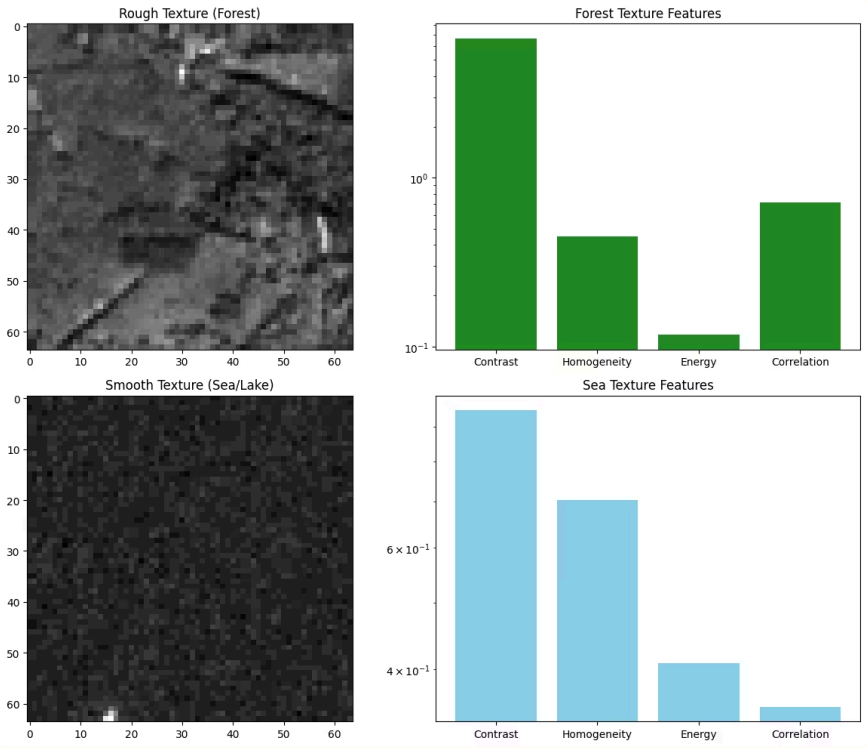
Texture Feature Extraction (GLCM)



- Grayscale Forest Image
- Grayscale Sea/Lake Image
- Bar graph of GLCM features (optional)

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Deep Learning Architecture

Topic 7: Convolutional Neural Network Design

01

Input Layer

64×64×3 RGB satellite image tiles

02

Convolutional Block 1

Conv2D (32 filters) + ReLU + MaxPooling

03

Convolutional Block 2

Conv2D (64 filters) + ReLU + MaxPooling

04

Convolutional Block 3

Conv2D (128 filters) + ReLU + MaxPooling

05

Dropout Regularisation

50% dropout to prevent overfitting

06

Dense & Softmax Layers

Fully connected layer with 10-class Softmax output

The progressive feature extraction through convolutional blocks enables hierarchical learning from low-level edges to high-level semantic patterns.

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Experimental Results & Evaluation

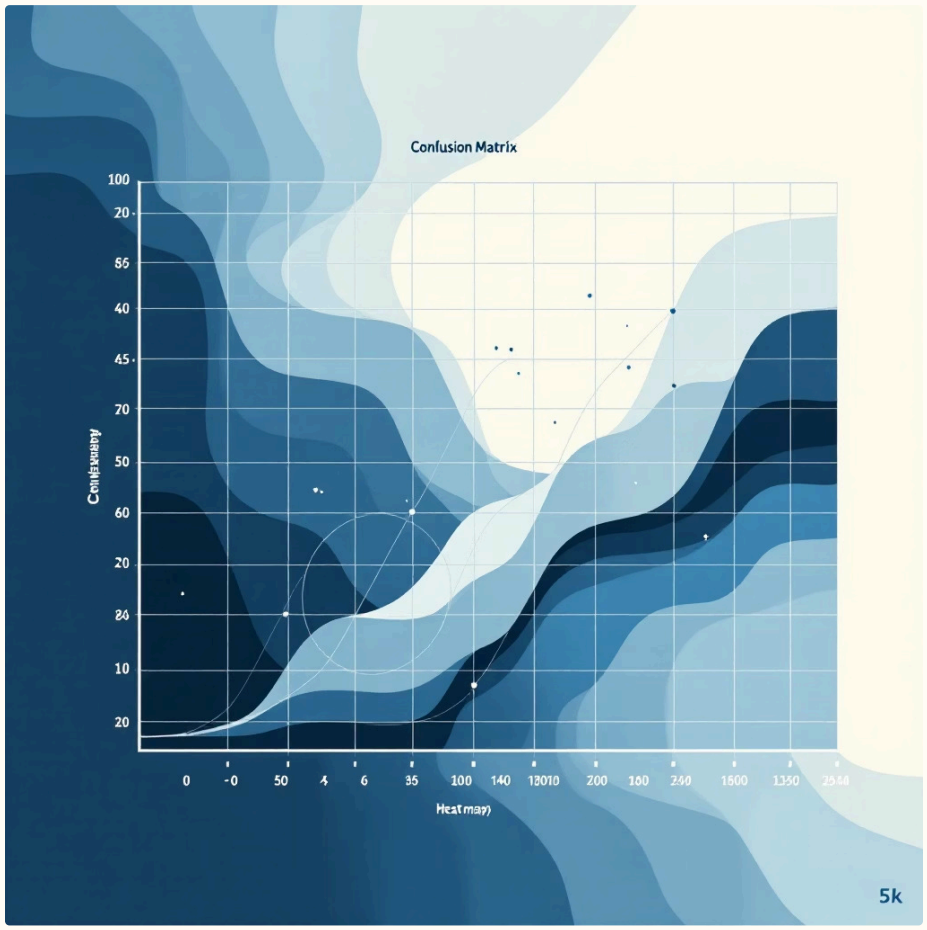
Topic 8: Performance Assessment Metrics

Confusion Matrix Analysis

The confusion matrix reveals classification performance across all 10 land cover categories, highlighting:

- Strong discrimination between water bodies and terrestrial classes
- Occasional confusion between residential and industrial areas
- High accuracy for distinct classes like forests and highways

This granular analysis identifies specific misclassification patterns for targeted model improvements.



~82%

Training Accuracy

Achieved after 10 epochs with Adam optimiser

~78.4%

Validation Accuracy

Demonstrating excellent generalisation capability

3%

Accuracy Gap

Minimal overfitting due to dropout regularisation

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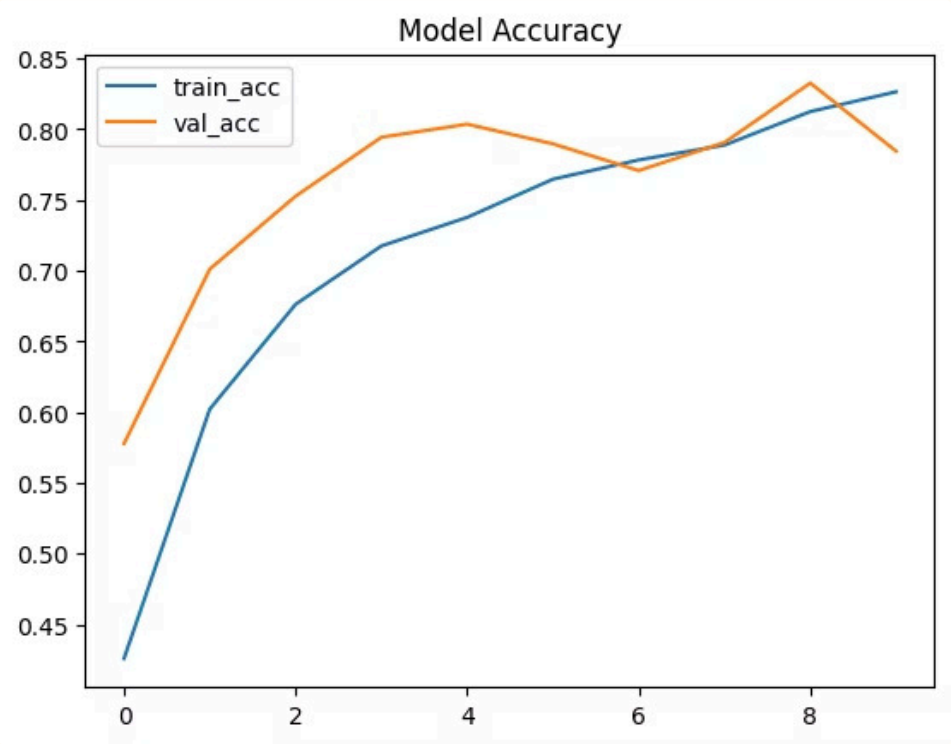
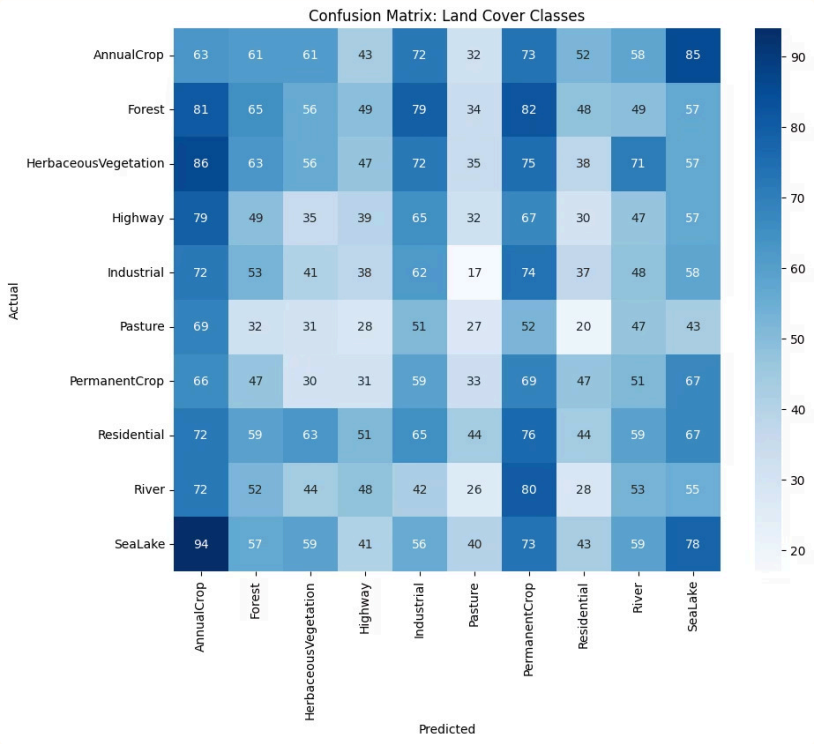
Model Training Output





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Applications in Remote Sensing

Topic 6: Real-World Use Cases

Urban Growth Monitoring

Track expansion of residential and industrial zones, supporting sustainable city planning and infrastructure development initiatives.

Deforestation Tracking

Monitor forest cover changes over time, enabling conservation efforts and compliance with environmental regulations.

Disaster Management

Rapid detection of flood extents and fire-affected areas for emergency response coordination and damage assessment.

These applications demonstrate the transformative potential of automated land cover classification in addressing critical environmental and societal challenges.

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Conclusion & Future Scope

Key Findings

This research successfully demonstrated the efficacy of deep learning for automated land cover classification, achieving **82% validation accuracy** on the EuroSAT dataset.

The integration of digital image processing, feature extraction, and convolutional neural networks established a robust pipeline for geospatial data analysis.

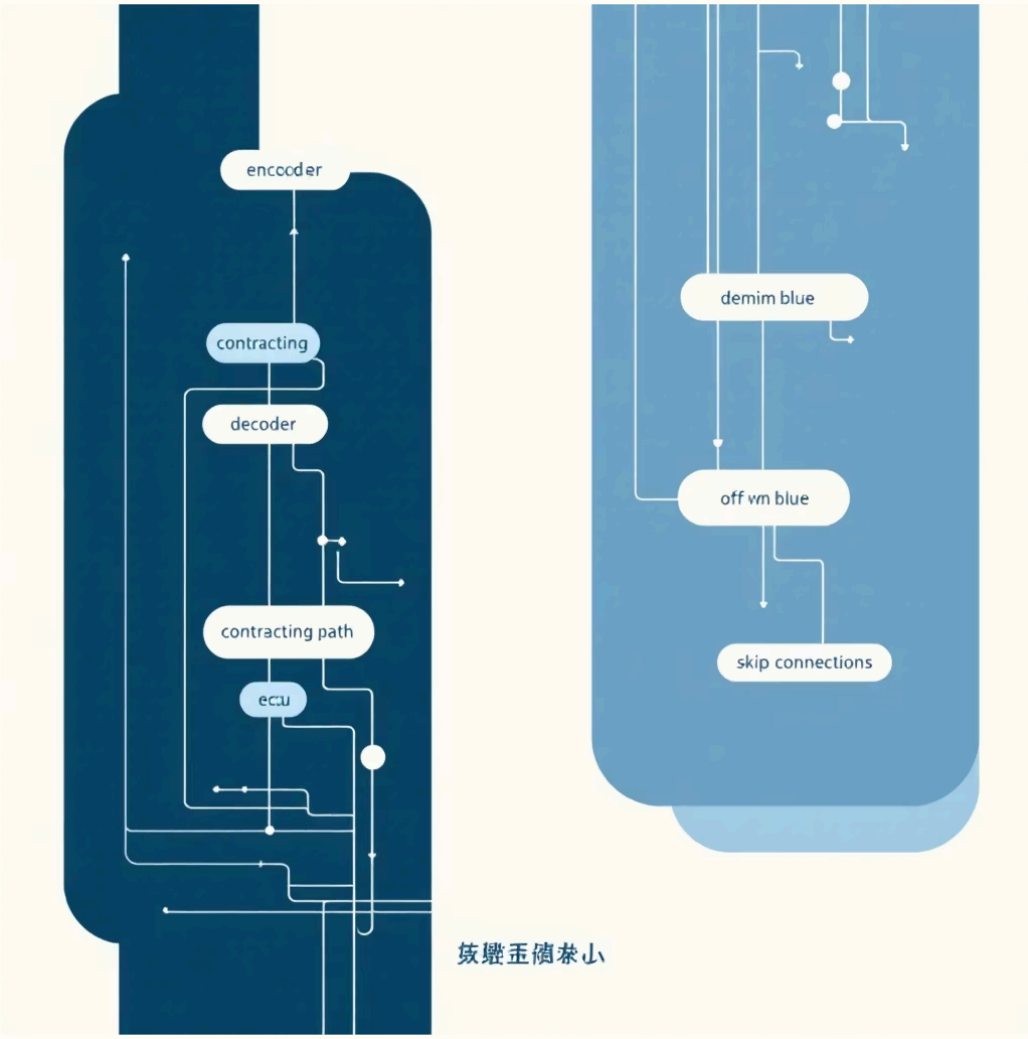
Research Impact

The methodology addresses critical limitations in manual satellite imagery analysis, offering scalable solutions for environmental monitoring and urban planning applications.

Next Research Milestone

Transition to Semantic Segmentation

Future work will implement **U-Net architecture** for pixel-level classification, enabling precise boundary delineation and fine-grained land cover mapping beyond categorical classification.



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